Madhiyougi Jain

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Education

UMass Amherst, Manning College of Information & Computer Sciences — Amherst, MA
B.S. Informatics (Data Science concentration) and Psychology — *Expected May 2027*

Projects

NYC Collisions: Injury Risk Watchlist | *Python, scikit-learn, DuckDB* | Live | GitHub — 2025 – 2026

- Gradient-boosted classifier scores a crash's injury risk from what was involved, not which site; on **121,970** held-out 2025–26 crashes, ROC-AUC **0.79** vs. **0.54** for a borough-hour base rate, Brier skill **+26.9%**.
- Ranked **1,093** intersections by the lower 95% bound of injury rate above their crash mix; in a backtest the ranking persists (Spearman **0.26**, $p = 0.0015$) but ties a no-model sort, with or without empirical-Bayes pooling.
- Scored **413,780** crashes with five rolling-forward fits so no crash is scored by a model that trained on it: **101** sites clear $z = 1.96$ where chance predicts 27, but only **2** survive Benjamini–Hochberg at 5% FDR.
- Weekly GitHub Actions job re-pulls a 30-day overlap from NYC Open Data, dedupes on `collision_id`, and rebuilds **637,256** crashes as Parquet; a freshness check tells a stalled source from a lagging build; **387** tests.

Hall Check: Dining-Hall Occupancy | *Python, PyTorch (YOLO), Docker, Linux* | Live | GitHub — 2026

- Person counter on four public UMass dining-hall cameras, live since Oct 2026: YOLO11n counts people inside an ROI polygon every 2 min, frames never leave memory, and CI fails the build on pixel-writing calls.
- YouTube blocks the streams from datacenter IPs (Fly.io), so the worker runs in Docker on a campus-network laptop at 376–447 MiB, after fixing a stream-format bug and a torch/torchvision mismatch the smoke test missed.
- Raised detector input from 640 to 1280 px after far-counter diners went unseen (21 vs. 7 people on one Berkshire frame), at 163 vs. 56 ms per frame on CPU.
- Monitoring separates failures: a daily drift alarm (40% and 4+ people across 6+ busy slots) flags a moved camera, and a health check flags any hall silent for 6 hours, which is the failure that blinds the drift check.

When Average Effects Lie: Uplift Modeling | *Python, scikit-learn, statsmodels* | Write-up | GitHub — 2026

- Trained T-, S-, and DR-learners on the **64,000**-customer Hillstrom randomized trial, picking by 5-fold cross-fitted Qini on training rows; picking on the reporting split would have quoted a men's-email Qini of 19.3, not **2.6**.
- Found what the women's email's **+4.5pp** average visit lift hid: **+7.3pp** for prior women's-merchandise buyers vs. **+1.1pp** for the rest (Qini **60.4**), with Benjamini–Hochberg control across the interaction tests.
- Set the send threshold at cost break-even (assumed \$2/visit, \$0.06/email): per 1,000 customers, targeting adds **\$16.95** (95% CI [\$2.99, \$30.60]) and sends **415** fewer emails. For the men's email the same analysis said send to everyone, because its targeting gain could not be told from zero.

Experience

Somneuro Lab, UMass Amherst — Amherst, MA
Undergraduate Research Assistant — *Sep 2026 – Present*

- Onboarded (CITI certified) to the lab's wrist-actigraphy (GGIR) and polysomnography data from studies of sleep and memory consolidation.

Spiritual Data — Remote
AI & Psychology Research Intern — *May 2025 – Aug 2025*

- Built an automated harness that benchmarked a proprietary model against baselines over repeated runs.
- Designed evaluation splits and error slices to find failure modes; wrote the report that informed model selection.

Technical Skills

Languages: Python, SQL, Java, TypeScript, R
ML: PyTorch, Hugging Face Transformers, scikit-learn, Ultralytics YOLO; gradient boosting, calibration, uplift modeling
Statistics: statsmodels; permutation tests, FDR control, empirical Bayes, bootstrap CIs
Data: pandas, DuckDB, PostgreSQL
Systems & Tools: Linux, Docker, GitHub Actions (CI/CD), Git